SQL Notes

What is SQL ?
SQL(Structured Query Language) is a standard Database language that is used to create, maintain and retrieve the data from relational databases like MySQL, Oracle, etc.

The USE command

The use command is used when there are multiple databases in the SQL and the user or programmer specifically wants to use a particular database.
In simple terms, the use statement selects a specific database and then performs operations on it using the inbuilt commands of SQL.
Syntax:

```
USE database_name;
```

Example:
```
USE candy;
```

The show tables Command

Select the specific database you want to view using the USE command

```
USE database_name;
```

Execute the SHOW TABLES command to list all non-temporary tables in the selected database:
Syntax:
```
SHOW TABLES
```
Example:
```
USE candy;
SHOW TABLES;
```

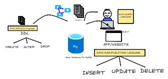
Describe Command

Syntax:
```
DESCRIBE table_name;
```

Example:
```
DESCRIBE movie;
```

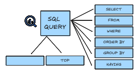
```
USE <db name>;
SHOW TABLES;
DESCRIBE<table-name>
```

Summary



Where

Filter your data based on condition
SYNTAX:
```
SELECT column1, column2, ...
      FROM table_name
      WHERE condition;
```



ORDER BY :

SORT YOUR DATA

SYNTAX :

```
SELECT column1, column2, ...
   FROM table_name
ORDER BY column1, column2, ... ASC|DESC;
```

EXAMPLE :
```
SELECT * FROM Products
   ORDER BY Price;
```

GROUP BY :

COMBINE ROWS WITH SAME VALUE

AGGREGATE A COLUMN WITH ANOTHER COLUMN

SYNTAX :
```
SELECT column_name(s)
   FROM table_name
   WHERE condition
GROUP BY column_name(s)
ORDER BY column_name(s);
```

EXAMPLE :

```
SELECT
    country,
    first_name,
    SUM(score) AS TOTAL_SCORE
FROM customers
GROUP BY country, first_name
```

DISTINCT :

REMOVE DUPLICATES (REPEATED VALUES)

EACH VALUE APPEARS ONLY ONCE

SYNTAX :
```
SELECT DISTINCT column1, column2, ...
      FROM table_name;
```

EXAMPLE :
```
SELECT DISTINCT country
   FROM customers
```

TOP :

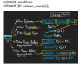
RESTRICT THE NUMBER OF ROWS RETURNED

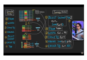
HAVING :

FILTERS DATA AFTER AGGREGATION

CAN BE USED ONLY WITH GROUP BY

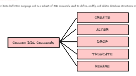
SYNTAX :
```
SELECT column_name(s)
FROM table_name
WHERE condition
GROUP BY column_name(s)
HAVING condition
ORDER BY column_name(s);
```





DATA DEFINITION LANGUAGE

DDL stands for Data Definition Language and is a subset of SQL commands used to define, modify, and delete database structures or schemas



CREATE: Used to create the database or its objects, such as tables, indexes, or views.
Example:
```
CREATE TABLE
Employees (ID INT PRIMARY KEY, Name VARCHAR(100));
```

EXAMPLE :

```
USE MyDatabase;

CREATE TABLE persons (
    id INT NOT NULL,
    person_name VARCHAR(50) NOT NULL,
    birth_date DATE,
    phone VARCHAR(15) NOT NULL,
    CONSTRAINT pk_persons PRIMARY KEY (id)
);
```

Used to modify the structure of an existing database object, for instance, by adding, modifying, or dropping columns.

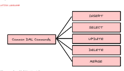
EXAMPLE

```
ALTER TABLE persons
ADD email VARCHAR(50) NOT NULL
SELECT * from persons
```

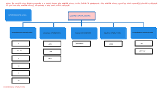
DROP : Used to delete an object (e.g., a table, index, or the entire database) and all its data and associated structures permanently.

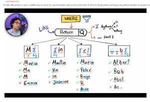
Example :
```
DROP TABLE persons
```

DATA MANIPULATION LANGUAGE



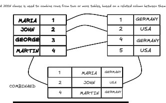
INSERT : Adds new rows (records) of data into a table.





JOINING

A JOIN clause is used to combine rows from two or more tables, based on a related column between them



NO JOINS

RETURN DATA FROMTABLES WITHOUT COMBINIG THEM

SYNTAX
```
SELECT * FROM A;
SELECT * FROM B;
```

INNER JOINS

RETURN ONLY MATCHING ROWS FROM BOTH TABLES

SYNTAX
```
SELECT column1, column2, ...
      FROM table1
   INNER JOIN table2
ON table1.column = table2.column;
```